

KMS Final Technology Stack Specification

Notice: The technologies listed below represent the mandatory and approved technology stack for the Knowledge Management System (KMS) project.

1. Frontend

- **Framework:** Next.js 15 (App Router)
- **Language:** TypeScript
- **Styling:** Tailwind CSS
- **UI Components:** Shadcn UI
- **Data Fetching & State:** TanStack Query (React Query)
- **Form Handling:** React Hook Form
- **Schema Validation:** Zod
- **Icons:** Lucide React

2. Backend

- **Runtime Environment:** Node.js
- **Framework:** Express.js
- **Language:** TypeScript
- **API Specification:** REST API
- **API Documentation:** Swagger (OpenAPI)

3. Database & ORM

- **Database:** PostgreSQL
- **ORM:** Prisma ORM

4. Authentication & Authorization

- **Authentication:** JWT & Refresh Token
- **Authorization:** Role-Based Access Control (RBAC)

- **Password Hashing:** bcrypt

5. File Storage

- **Media & File Storage:** ImageKit

6. Security

- **Security Headers:** Helmet
- **Origin Control:** CORS
- **Throttling:** Rate Limiting
- **Input Validation:** Zod

7. Caching & Logging

- **In-Memory Cache:** Redis
- **Logger:** Pino

8. Monitoring & Error Tracking

- **Metrics Collection:** Prometheus
- **Visualization:** Grafana
- **Error Tracking:** Sentry

9. Testing

- **Unit & Integration Testing:** Jest
- **API Testing:** Supertest

10. DevOps & Infrastructure

- **Containerization:** Docker
- **Reverse Proxy:** Nginx
- **CI/CD Automation:** GitHub Actions
- **Host Environment:** Ubuntu Server / VPS